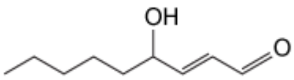
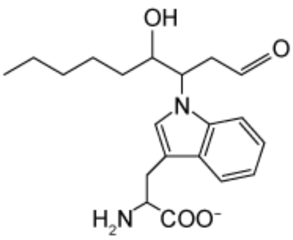
4-Hydroxy-2-nonenal (4HNE), shown below, interacts with proteins by forming covalent bonds with the nucleophilic side chains of amino acids.



4HNE structure

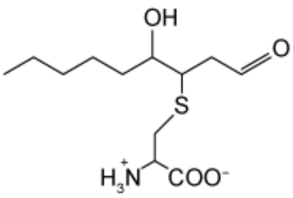
If a protein were incubated with 4HNE and then degraded by proteases, which of the following could represent the structure and charge of an amino acid bound to 4HNE at pH 7?

○ A.



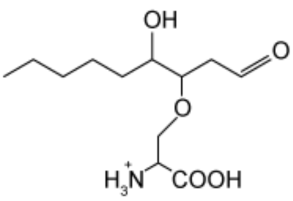
[12%]

✔ ● B.



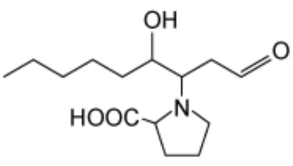
[56%]

○ C.



[23%]

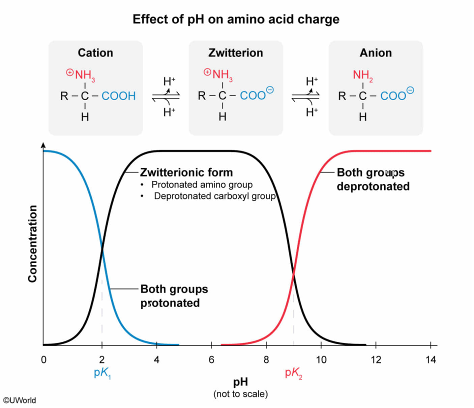
○ D.



[7%]

Correct
55% answered correctly

Explanation:



All amino acids contain a carboxyl group and an amino group in their backbones that can be protonated or deprotonated depending on the pH to which they are exposed. The carboxyl group has a pK_a of approximately 2 and typically acts as a proton donor; the amino group has a pK_a of about 9 and generally acts as a proton acceptor. At or near physiological pH (7.4), amino acid backbones exist as **zwitterions**, molecules that have both a positive and a negative charge, which cancel each other; the carboxyl group is deprotonated (–COO^-) and carries a negative charge, and the amino group is fully protonated (–NH_3^+) and has a positive charge.

According to the question, 4-hydroxy-2-nonenal (4HNE) forms covalent bonds with nucleophilic amino acids, which contain side chains with atoms that can donate electrons to form covalent bonds. The question states that this process occurs at pH 7. This pH is higher than the pK_a of the carboxyl group but lower than the pK_a of the amino group; therefore, the amino acid would exist predominantly as a zwitterion. Of the available answer choices, only the complex attached to cysteine shows a nucleophilic amino acid with the appropriate zwitterionic backbone.

(Choices A and D) Nonpolar amino acids such as tryptophan and proline do not contain nucleophilic side chains. In addition, the backbone amino group would be protonated at pH 7 in tryptophan and proline. The backbone carboxyl group would be deprotonated at pH 7 in proline.

(Choice C) This diagram shows serine with its N- and C-termini fully protonated. Although serine is a nucleophilic amino acid and could attack 4HNE, at pH 7 the C-terminus should be *deprotonated*, not protonated.

Educational objective:

At or near physiological pH (7.4), amino acid backbones predominantly exist as overall neutral molecules referred to as zwitterions. They have a net charge of zero because the carboxyl group (pK_a of 2) is deprotonated (existing as –COO^-) while the amino group (pK_a of 9) is protonated (existing as –NH_3^+). Nucleophilic amino acids contain side chains with atoms that donate electrons when they are deprotonated.

Time spent:0 sec
Subject:
Biochemistry

QID:402126
System:
Amino Acids and Proteins